Success Factors in Systems Interoperability Projects: A Preliminary Exploratory Study

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Abstract. Interoperability between information systems is fundamental for data sharing and process integration, but its implementation involves technological, organizational, human, and managerial challenges. This paper presents a preliminary exploratory study that surveys the success factors reported in the literature on interoperability projects. It is the first stage of a broader research project that aims to validate these factors through a case study at the Mato Grosso State Department of Health; this stage is necessary because it provides the initial framework that will guide data collection and analysis in that case study. Based on 19 studies, five groups of factors were identified: management, leadership, and governance; change management and capacity building; strategic and institutional alignment; infrastructure and technology architecture; and data quality, security, and standardization. The results are preliminary: no formal selection protocol was applied, and the case study is left for future work.
Keywords: System Interoperability, Success Factors, Enterprise Application Integration.

Resumo. A interoperabilidade entre sistemas de informação é fundamental para o compartilhamento de dados e a integração de processos, mas sua implementação envolve desafios tecnológicos, organizacionais, humanos e gerenciais. Este artigo apresenta um estudo exploratório preliminar que levanta os fatores de sucesso relatados na literatura sobre projetos de interoperabilidade. Trata-se da primeira etapa de um projeto de pesquisa mais amplo, que visa validar esses fatores por meio de um estudo de caso na Secretaria de Estado de Saúde de Mato Grosso; essa etapa é necessária, pois fornece a estrutura inicial que norteará a coleta e a análise de dados no referido estudo de caso. Com base em 19 estudos, foram identificados cinco grupos de fatores: gestão, liderança e governança; gestão de mudanças e capacitação; alinhamento estratégico e institucional; infraestrutura e arquitetura tecnológica; e qualidade, segurança e padronização de dados. Os resultados são preliminares: não foi aplicado um protocolo formal de seleção, e o estudo de caso fica reservado para trabalhos futuros.
Palavras-chave: Interoperabilidade de Sistemas, Fatores de Sucesso, Integração de Aplicações Corporativas.

1. Introduction

Digital transformation drives organizations to identify critical success factors for technology adoption and integration [Pasqual et al. 2023]. Integrating IT operations and cross-departmental processes is challenging for delivering integrated public services [Kamal et al. 2013]. Developing interoperability in enterprise networks is likewise challenging, with maturity models and ontologies proposed to diagnose barriers [Guédria 2014].

This study is motivated by the need to identify and understand the factors that contribute to the success of interoperability projects, supporting risk mitigation, resource optimization, and more effective project planning. In healthcare, the lack of interoperability between systems reduces the quality of care and wastes resources [Torab-Miandoab et al. 2023].

This paper represents the first stage of a broader research project to validate interoperability success factors in public healthcare through a case study at the Mato Grosso State Department of Health. In Brazil, recent findings show that integrating primary and hospital health data across municipal and federal platforms is feasible even within a complex federative context [Fernandez et al. 2025].

2. Methodology

This is an exploratory, qualitative study, conducted as the first stage of the broader project described above. Its goal is to gather the success factors mentioned in the literature, not to review it exhaustively.

The literature was searched in Google Scholar (broad coverage across publishers and areas) and the ACM Digital Library (computing, software architecture, and information systems), serving as starting points for this preliminary stage; other databases are planned for future stages. The initial set was identified using the terms ("critical success factors" OR "CSF") AND ("system interoperability" OR "enterprise application integration") AND ("data sharing" OR "process integration"). In October 2026, this set was complemented with recent studies on interoperability in healthcare and the public sector, identified through web searches and individually checked in Google Scholar to improve temporal and thematic coverage.

The analyzed set was defined by selecting peer-reviewed English journal and conference articles (2006–2025) addressing success factors, challenges, or enablers in interoperability or related domains (ERP, enterprise architecture, and digital transformation), while excluding duplicates and non-peer-reviewed material. 19 studies were selected (14 from 2016 onwards; 12 from 2023 onwards). Eight address interoperability or application integration directly, whereas eleven provide complementary evidence from adjacent domains—such as digital transformation [Pasqual et al. 2023], digital procurement [Luo et al. 2023], water-industry IoT [Narang et al. 2025], and blockchain recordkeeping [Wang and Yang 2021]—deliberately included to capture cross-cutting digital transformation factors that can inform the healthcare model.

For each study, the success factors, challenges, and enabling conditions reported by the authors were listed. Factors with similar meaning were merged into five groups, each named after the dominant theme of its factors. The main factors of each group are shown in Table 1. As no formal protocol was followed and the grouping was not independently validated, the set should not be seen as complete or free of selection bias (see Section 4).

3. Preliminary Results

The studies were organized into five groups of success factors, shown in Table 1 together with the main factors of each group. Studies marked with an asterisk (*) address interoperability or application integration directly; the others come from related contexts and are treated as complementary evidence. The number of studies per group does not indicate its relative importance. Figure 1 summarizes the groups as the initial framework to be validated in the case study.

Table 1. Groups of success factors, main factors, and related studies

Group of success factors	Main factors reported	Related studies (* = directly on interoperability or integration)
Management, leadership, and governance	Top management support; organizational anchoring; economic and political factors; strategic and policy challenges; stakeholder collaboration; regulatory compliance; leadership and vision	[Hjort-Madsen 2006*, Lange et al. 2016, Hillmann et al. 2025, Sixpence and Vasconcelos 2024*, Rodriguez Müller et al. 2024*, Fernandez et al. 2025*, Leyh et al. 2025, Pasqual et al. 2023]
Change management and capacity building	Change management; user training and participation; team expertise	[Zouine and Fenies 2014, Butarbutar et al. 2023, Hillmann et al. 2025, Leyh et al. 2025]
Strategic and institutional alignment	IT/business alignment; organizational fit	[Gericke et al. 2010*, Zouine and Fenies 2014, Leyh et al. 2025]
Infrastructure and technology architecture	Architecture management; infrastructure quality; interoperability layers and barriers; API design patterns; flexible IT infrastructure; federated platforms	[Gericke et al. 2010*, Lange et al. 2016, Bogner et al. 2024, Guédria 2014*, Kamal et al. 2013*, Sixpence and Vasconcelos 2024*, Rodriguez Müller et al. 2024*, Fernandez et al. 2025*]
Data quality, security, and standardization	Information quality; semantic interaction; trustworthy recordkeeping; risks of digitalizing health data; digital procurement and IoT-based decisions	[Zouine and Fenies 2014, Narang et al. 2025, Wang and Yang 2021, Ntshangase et al. 2025, Luo et al. 2023, Torab-Miandoab et al. 2023*]

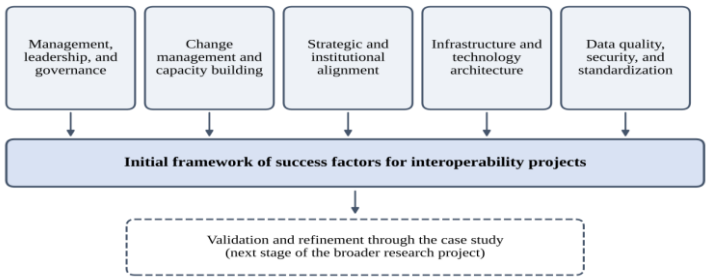


Figure 1. Initial framework of success factors

3.1 Management, leadership, and governance

Executive support and organizational anchoring are recurrently cited as critical determinants for mitigating project failures in integration and enterprise architecture initiatives [Lange et al. 2016, Hillmann et al. 2025, Leyh et al. 2025]. The literature shows that interoperability extends beyond technical connectivity to encompass complex political, economic, and regulatory dimensions [Hjort-Madsen 2006, Sixpence and Vasconcelos 2024], where project success hinges upon strategic alignment and robust inter-institutional stakeholder collaboration [Rodriguez Müller et al. 2024, Fernandez et al. 2025]. In digital transformation, leadership, strategy, and vision weigh more than technology [Pasqual et al. 2023].

3.2 Change management and capacity building

In the meta-analysis of 32 ERP studies by [Zouine and Fenies 2014], individual impact (including user involvement and training) had the strongest relation with ERP success. [Butarbutar et al. 2023] and [Hillmann et al. 2025] point to insufficient training and expertise as challenges, and user training is among the top three factors in [Leyh et al. 2025].

3.3 Strategic and institutional alignment

[Gericke et al. 2010] include IT/business alignment among application integration success factors, and organizational fit is the second most important factor in [Leyh et al. 2025]; organizational impact, which includes fit, moderately relates to ERP success in [Zouine and Fenies 2014].

3.4 Infrastructure and technology architecture

Infrastructure quality and architecture management are technological enablers of application integration [Gericke et al. 2010, Lange et al. 2016]. Integration technologies work mainly as back-office mechanisms supporting flexible IT architectures [Kamal et al. 2013], while API design patterns and maturity models address understandability and integration barriers [Bogner et al. 2024, Guédria 2014]. Effective data exchange requires technical, semantic, and organizational layers [Sixpence and Vasconcelos 2024, Rodriguez Müller et al. 2024], as in federated health platforms [Fernandez et al. 2025].

3.5 Data quality, security, and standardization

Information quality strongly relates to ERP success [Zouine and Fenies 2014], and semantic interaction is indicated as the best choice to guarantee health data exchange [Torab-Miandoab et al. 2023]. [Wang and Yang 2021], [Ntshangase et al. 2025], [Luo et al. 2023], and [Narang et al. 2025] address recordkeeping, digital health risks, digital procurement, and IoT-based decisions, respectively.

3.6 Recurrence over time and contribution

Top management support appears from 2014 [Zouine and Fenies 2014] to 2025 [Hillmann et al. 2025, Leyh et al. 2025], and strategic alignment from 2010 [Gericke et al. 2010] to 2025 [Leyh et al. 2025], which suggests that these factors recur over time, although older studies reflect earlier technologies. The five groups overlap with dimensions already discussed in enterprise architecture management and ERP success studies, which also report top management support, organizational anchoring, team expertise, and user training [Lange et al. 2016, Hillmann et al. 2025, Leyh et al. 2025]. The contribution is therefore a synthesis, not new dimensions: it reorganizes factors from enterprise architecture, ERP, and application integration studies around interoperability, separates evidence by proximity, and offers the groups as a starting point for validation in public health.

4. Conclusion

This preliminary exploratory study surveyed the success factors of interoperability projects reported in the literature and organized them into five groups. The results indicate that success goes beyond technical connectivity between systems and requires well-structured governance and constant alignment between IT and the institution's operations.

The contribution of this stage is the initial framework (Figure 1) that will guide the case study.

In the next stage, each group will serve as an analysis dimension of the case study, with indicators such as management support, training coverage, fit with institutional goals, standards and system compatibility, and semantic standards and security controls, to be refined with empirical evidence.

This study has limitations: the search relied on two databases and complementary web searches, no systematic selection protocol was used, the grouping was not independently validated, and some studies address related topics (ERP, enterprise architecture) rather than interoperability itself. Future work includes expanding the search (IEEE Xplore, Scopus, Web of Science) with a structured protocol (PRISMA or Kitchenham), carrying out the case study to validate the groups, using Brazilian experiences [Fernandez et al. 2025] as a reference, and investigating the role of artificial intelligence and automation in interoperability.

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